

A Stacked Embedded DRAM Array for LPDDR4/4X using Hybrid Bonding 3D Integration with 34GB/s/1Gb 0.88pJ/b Logic-to-Memory Interface

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Abstract—Increasing demand for DRAM scaling and high-bandwidth has driven DRAM technology to 3D/2.5D integration. With the innovative Hybrid Bonding technology, a new Stacked Embedded DRAM (SEDRAM) architecture was developed on LPDDR4/4X product. In this SEDRAM, a DRAM array wafer and logic wafer were fabricated separately and then face-to-face fusion connected through ultra-high-density, low-resistance Hybrid Bonding. By separating the control, I/O, DFT and periphery circuits to a logic die, SEDRAM offers a novel approach to DRAM product development and accomplishes an extremely high-bandwidth logic-to-memory interface speed of 34GBps per 1Gb with low power consumption as low as 0.88pJ/bit.

I. INTRODUCTION

DRAM has been chosen as main memory due to its relatively low latency and high density. DRAM scaling has enabled lower cost per bit by reducing cell size for decades. Unfortunately, further scaling of DRAM has become costly due to manufacturing complexity/cost and difficult to keep up with the continuous scaling trends. Meanwhile DRAM bandwidth has fallen behind the requirements of logic computing.

Nowadays many applications, such as high performance computing and video processing need more and more bandwidth. As a solution, 3D/2.5D integration, especially HBM, receives great attention from both academia and industry [1]. HBM uses silicon interposer and microbump to increase the number of I/O connection of logic-to-memory interface to provide high bandwidth at high data rate. However, further increase of the per-pin data rate requires complex and power hungry PHY circuitry in both HBM and logic sides. It's more effective to increase the connection density for higher bandwidth. Besides, the memory I/O power becomes more dominant in the overall system power consumption. TSV and interposer connections have large resistance and capacitance, causing high power consumption. It's advantageous to use either wide but low speed I/O or low-resistance and low-capacitance I/O at the memory interface.

To mitigate the aforementioned issues, a new advanced hybrid bonding technology is introduced to our DRAM product

development [2]. In this proposed SEDRAM, the periphery circuits, including control, I/O and DFT, are separated to a logic die and stacked above an array die by hybrid bonding, which uses Cu to Cu direct fusion bond. The DFT block is designed as an IP in the logic die to perform BIS (Built-In Self-Repair) for the array die. Since SEDRAM utilizes fully independent process technologies for array and logic die, the two dies can be designed and optimized for their own best performance respectively. This approach also opens the possibilities for special interface DRAM development and high density embedded DRAM solutions by incorporating of innovative functionalities in logic die. Moreover, hybrid bonding via density of 110,000/mm² with 3um pitch has been successfully demonstrated in this work. This is a few hundred times denser than the microbump approach used in HBM. As an extension of Back End Of Line (BEOL) interconnect process, hybrid bonding via resistance of less than 0.5ohm/ea was obtained with parasitic capacitance much lower than the microbump and TSV counterparts. As a result, the logic-to-memory interface power consumption can be reduced by 40% as well.

Finally we have successfully developed a JEDEC fully complied 4Gb LPDDR4/4X based on SEDRAM, where the logic-to-memory interface bandwidth is 34GBps per 1Gb and power efficiency is 0.88pJ/bit. Since the 1Gb memory unit in SEDRAM is independent, we can easily extend the bandwidth to several TBps while retain power efficiency by placing more memory units.

II. STRUCTURE OF SEDRAM

Figure 1 shows the structure of the SEDRAM chip. At the top, there is a logic die facing downward, and at the bottom there is a DRAM die. All control logics and I/O circuits are on the logic die. The array die is built up with a repeatable memory unit of 1Gb, which consists of 8 array blocks of 128Mb and independent on-chip power system. Each 128Mb array block is an independent memory channel and has 128 I/Os. Compared to HBM, the total I/O bus between DRAM die and logic die is wider. Even the data rate runs as low as 266MHz, the total bandwidth of our 4Gb SEDRAM can reach 136GBps.

Hybrid bonding pitch size is 3um, on the contrary, the pitch of microbump was about 50um[3,4] and the pitch of TSV was

about 6 μ m[5,6]. Compared to HBM using microbump and TSV technologies, SEDRAM using Hybrid bonding technology can reach a maximum via density of 110,000/mm². Figure 2 illustrates the key process for stacking the logic and array wafers. After the top metal layer is fabricated, each logic/array wafer is passed to hybrid bonding process. Planarization, photography and dry etch processes are implemented to form top vias (LTVIA) and bottom vias(LBVIA) first. Then the logic wafer is turned over and bonded face-to-face with the array wafer. After that, we thin the Si substrate of the logic wafer to about 3 μ m, and finally open the PAD out window from backside. It's obvious that the form factor of hybrid bonding is much smaller than microbump, TSV and wiring lines in HBM.

Figure 3 shows the SEDRAM cross sectional TEM image. The logic wafer (top) and array wafer (bottom) are bonded face to face by Hybrid bonding. The dotted lines are the interface between two wafers. We compared SEDRAM using hybrid bonding with HBM [3,4] and stacked sensor with DRAM[5,6] of recently published papers in Table-1. By stacking the logic wafer and array wafer and developing a fine pitch hybrid bonding, it is possible to achieve much higher bandwidth with much reduced I/O power consumption simultaneously.

III. EXPERIMENTAL RESULTS

A. Hybrid bonding test results

Hybrid bonding reliability tests include JEDEC standard High Temperature Stress (HTS), Temperature Humidity Stress (THS), Temperature Cycling (T/C) and Electro-Migration (EM). HTS, THS and T/C tests monitor resistance and leakage degradation after environmental stress. While EM test simulates the current tolerance of the bonding via and interface.

Figure 4 shows the cumulative failure plot of hybrid bonding test after HTS 1000hrs. The horizontal axis illustrates resistance shift rate and leakage, and the vertical axis is the cumulative failure rate. The maximum resistance shift was suppressed to -6.95%. When 29V voltage was applied between hybrid bonding chain and substrate, the leakage current was as low as 18pA with negligible variation.

During THS, the test structure was baked and then followed by the humidity environment stress. After 1000hrs stress, the plot charts in Figure 5 show that resistance shift was as low as -1.26%. Except for some structures at wafer edge, the leakage degradation was negligibly small. It indicated the hybrid bonding process was strong to encounter humidity attack.

The thermal cycle (T/C) was used to evaluate the hybrid bonding structure's fatigue properties. The environment temperature ramp up and down in each cycle. After 500 cycles, the resistance shift was very convergent and the max shift was -2.52%. Additionally, the leakage was on a very low level of 16.9pA, shown in Figure 6.

As hybrid bonding structure can be considered as an equivalent BEOL via connection, the EM characterization was generally indicated as the metal via migration endurance under electric current stress. According to different current directions, upstream structure (LHDU: from array to logic) and downstream structure (LHBD: from logic to array) were evaluated. The lifetime were 2.0E+05 and 3.3E+03 years for upstream and downstream respectively, as shown in Figure 7.

B. SEDRAM test results

A 4Gb LPDDR4/4X product based on SEDRAM technology is fabricated to check the interface data rate and other potential risk introduced by hybrid bonding. The SEDRAM chip is complying with the LPDDR4/4X spec.

Figure 8 shows the 12-inch wafer, layout and performance summary of SEDRAM chip. The array wafer was fabricated in 25nm 3-metal DRAM process, while the logic wafer was from 55nm 7-metal logic process. There were totally more than 64K hybrid bonding vias per chip. The chip was 4Gb in capacity and 66.3 mm² in die size. All the I/O circuits were integrated in logic wafer and runs at 4266Mbps.

Figure 9 shows the SEDRAM access time shmoo plot result of tCK=0.56ns, which was the highest frequency of CP tester. For FT test, the DRAM components passed 4266Mbps for speed bin sorting in multiple cases, including HT(95C), HV(VDD1=1.2v, VDD1=2v) and LV (VDD2=1.05v, VDD1=1.65v). Because the SEDRAM chip was LPDDR4/4X of 16 pre-fetch, 4266Mbps of test speed represented that the data rate of hybrid bonding interface was 266MHz.

Figure 10 shows the DRAM retention time test result. The retention time of SEDRAM at 95C met target of 96ms. HBM with TSV and microbump usually has the adverse temperature rising issue, which degrades retention time. The SEDRAM with hybrid bonding can lower the interconnection thermal resistance and works at lower frequency. As a result, the retention time of SEDRAM can be equivalent to a reference LPDDR4/4X chip from conventional DRAM process.

IV. CONCLUSION

We introduced hybrid bonding technology to obtain 3D interconnects of excellent electrical and reliability results, and developed a 4Gb LPDDR4/4X for mass production. It consists of a 4 $\times$ 1Gb DRAM die, a logic die and more than 64K hybrid bonding vias. At the logic-to-memory interface of SEDRAM, 34GB/s per 1Gb bandwidth with power efficiency of 0.88pJ/bit has been successfully demonstrated. It is clearly advantageous to adopt hybrid bonding 3D integration to increase I/O connect density and reduce I/O power consumption between DRAM and logic. We believe that the feasibility shown by SEDRAM not only offers a new approach for conventional DRAM development but also enables a novel solution to special interface DRAM and embedded DRAM for bandwidth and density hungry applications.

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Fig. 1. Diagram of SEDRAM(4Gb)

Fig. 2. Process flow of SEDRAM with hybrid bonding.

Fig.3. SEDRAM cross sectional TEM images.

(a) resistance shift

(b) leakage

Fig 4. Cumulative plot of hybrid bonding after HTS 1000hrs

(a) resistance shift

(a) resistance shift

(a)LHBU

(b) leakage

(b) leakage

(b) LHBD

Fig 5. Cumulative plot of hybrid bonding after THS1000hrs

Fig 6. Cumulative plot of hybrid bonding after T/C 500cyc

Fig 7. Cumulative plot of hybrid bonding EM
Time-to-fail

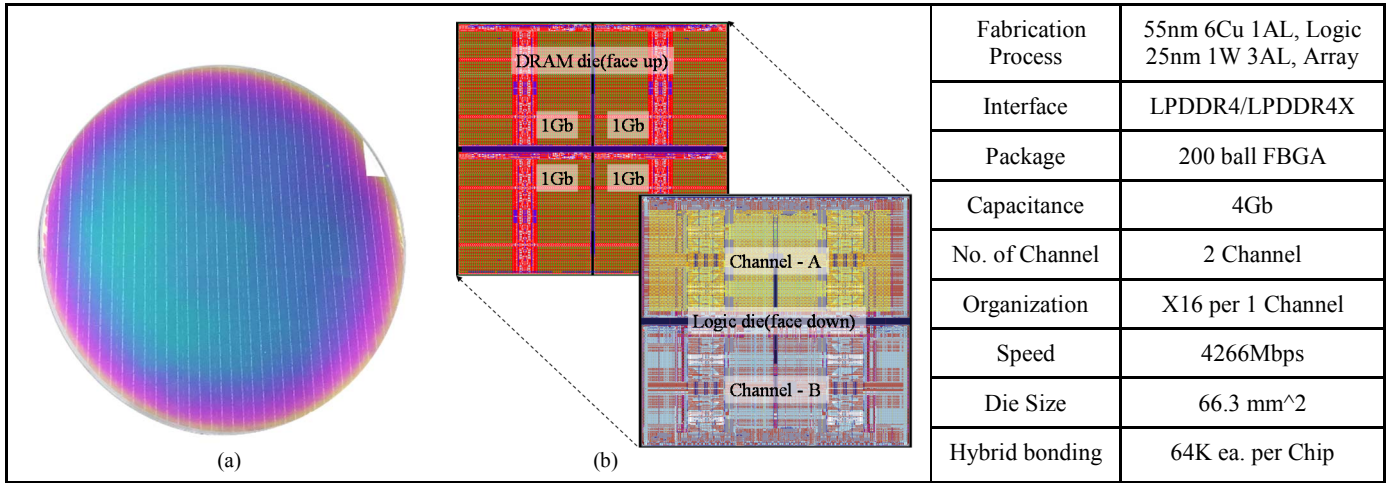


Fig 8. Wafer(a), layout(b) and performance summary of LPDDR4/4X based on SEDRAM

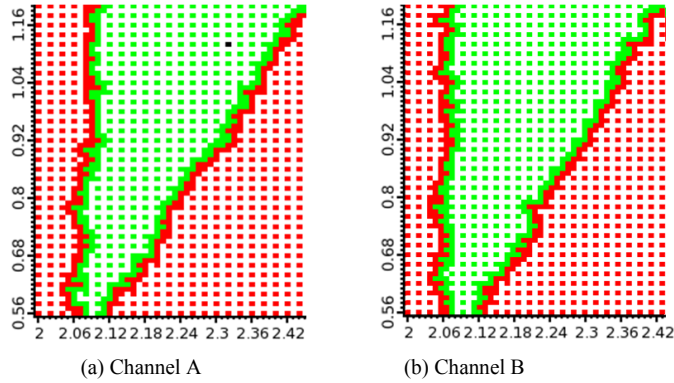


Fig 9. SEDRAM Access time shmoo result

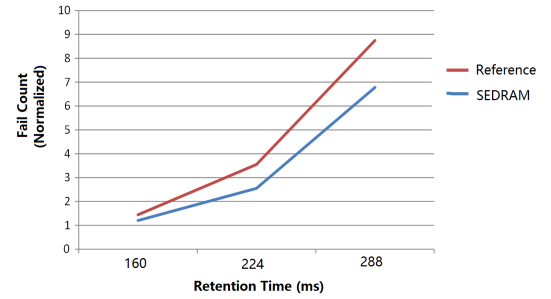


Fig 10. SEDRAM retention time result

	ISSCC14[3]	ISSCC20[4]	ISSCC17[5], IEDM17[6]	this proposal
Structure diagram				
Connection,pitch(um)	microbump, 48 x 55, interposer,		TSV,6.3 x 6.3, no interposer	Hybrid Bonding, 3 x 3, no interposer
Connection length	~5mm, microbump+wiring		~10um, TSV+wiring	2um, via thickness
PHY needed	Yes		No	No
Energy efficiency(pJ/b)	~1.5[2]		N/A	0.88
Total Density	8Gb	128Gb	1Gb	4Gb
# of Stack dies	4	8	1	1
Density per die	2Gb	16Gb	1Gb	4Gb
# of Channel	8	8x2	4	32
Data bus width	1024(128/ch)	1024(64/ch)	512(128/ch)	4096(128/ch)
Data rate per pin(Mbps/pin)	1000	4000	200	266
Bandwidth(GBps)	128	512	12.8	136
Bandwidth per die(GBps)	32	64	12.8	136

Table 1. Summary of SEDRAM and HBM